THE ETA INVARIANT IS MULTIPLICATIVE IN THE NONZERODIVISOR: ETA_F ETA_G EQUALS ETA_FG

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'We omit the details.'; that sentence is kept verbatim and marks the one step left to the reader.

Allowed standard prerequisites (editorial).

- The eta invariant eta_f M^bullet of a complex with respect to a nonzerodivisor f. fixed earlier in the same section and used on both sides of the statement at source line 27843. Fixes the two constructions being compared and the submodules they produce in each degree.
- Canonical identification of the iterated localization (M^i_g)_f with M^i_{fg}. used at proof line 27847. Supplies the common ambient module in which the two submodules are compared.

Lemma 0.2. *Let A be a ring and let $f, g \in A$ be nonzerodivisors. Let $M^\bullet$ be a complex of A-modules such that fg is a nonzerodivisor on all M^i. Then $\eta_f \eta_g M^\bullet = \eta_{fg} M^\bullet$.*

Proof. The statement means that in degree i we obtain the same submodule of the localization $M^i_{fg} = (M^i_g)_f$. We omit the details. □

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